



Using the OxO mapping service

Finding, reading, and trusting mappings between ontology terms.

A guide to the OxO2 web interface · written 2026-07-08 against branch `dev`, commit `e2efb77`. If a control described here is missing from your instance, the instance is older than this document.

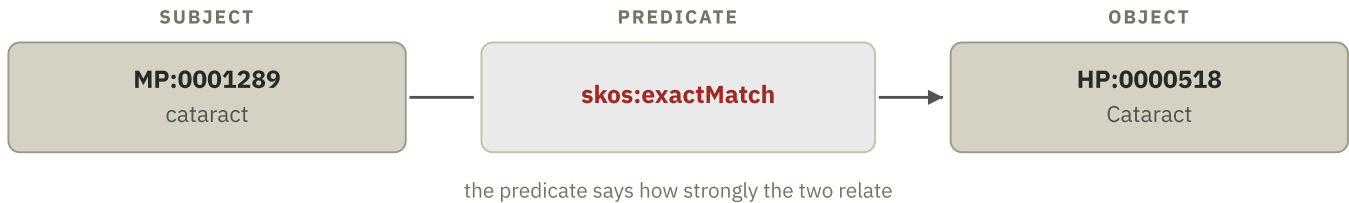
OxO answers one question: *this term over here — what is it called over there?* You give it a term, it gives you the mappings other people have already published, tells you who published each one and how confident they were, and gets out of the way.

CONTENTS

1. What OxO does
2. The screens
3. Searching
4. Advanced search
5. Reading the results
6. How “Best match” ranks
7. The mapping detail page
8. Explaining an inferred mapping
9. Inferred mapping sets
10. Glossary
11. Things that will surprise you

1 What OxO does

A **mapping** is a statement that one term means (roughly) the same thing as another. OxO stores mappings in the SSSOM format, where every mapping is a sentence with three parts:



Every row in an OxO result is one of these sentences.

OxO holds mappings from two kinds of source, and the distinction matters when you decide what to trust:

- **Ontologies.** Cross-references an ontology asserts about its own terms — the `xref` s the ontology maintainers put there themselves.
- **Curated sets.** Mappings published as standalone SSSOM files, such as EVORA or Mapping Commons, usually the output of a deliberate curation effort.

On top of both, OxO derives new mappings by chaining existing ones: if `A exactMatch B` and `B exactMatch C`, then `A exactMatch C`. These are **inferred** mappings, marked **SSSOM inference** throughout the interface, as opposed to **Asserted** ones that somebody actually wrote down. An inferred mapping is a real answer, but it is a derived one, and OxO can always show you the chain it came from (§ 8).

2 The screens

Screen	Address	What it is for
Home	<code>/</code>	The search form, plus links to About, related tools, and the issue tracker.
Results	<code>/search/<terms></code>	The search form again (so you can refine without going back), then the results table.
Mapping details	<code>/mapping/<id></code>	Everything SSSOM records about one mapping, and — if it is inferred — the graph explaining it.
Inferred sets	<code>/inferences</code>	Browse the mappings OxO derived, as a mapping set in its own right.
About	<code>/about</code>	What OxO is and who maintains it.

The search form on the Home page and the one above the results are the same component. Refining a search never means navigating backwards.

3 Searching

The **Search** tab asks one question and puts everything else out of the way. The **Advanced** tab (§ 4) is a different tool for a different job.

3.1 One term, or many

Single term

Multiple terms

Which term do you want to map?

A label, CURIE or IRI — e.g. cataract, MP:0001289

Single-term mode. Press Enter to search.

Single term gives you one input box; Enter searches. **Multiple terms** swaps it for a text area — one term per line — plus a file drop and a running count of how many terms you have. In multiple-term mode Enter starts a new line, so use the **Search** button.

Both modes share the same **Example...** link, which fills the box with something that works. Use it once and you will never need this section again.

3.2 What you can type

OxO recognises three kinds of term, and you may mix them freely in one search:

Kind	Example	How it is matched
Label	cataract	Against term labels. Exactly, ignoring case, unless you change it (§ 3.7).
CURIE	MP:0001289	Exact identifier lookup. Always exact, whatever the label-matching setting says.
IRI	http://purl.obolibrary.org/obo/UBERON_0002107	Exact IRI lookup. Also always exact.

Terms are separated by newlines or commas, and surrounding whitespace is trimmed.

WHY LABEL SEARCH IS EXACT BY DEFAULT

A partial search for `cataract` also returns “*congenital cataract*”, “*cataract surgery*”, and several hundred others — which buries the term you actually meant. Exact-but-case-insensitive is almost always what you want. Switch to **Partial match** under *More options* when you are exploring rather than looking something up.

3.3 Loading a list of terms from a file

In multiple-term mode you can drop a `.txt`, `.csv` or `.tsv` file (up to 1 MB) onto the dashed box, or click to pick one. The file is read **in your browser** — nothing is uploaded anywhere — and its terms are *appended to the text area*.

That last detail is deliberate. The text area is always the single, visible, editable list of what will be searched. A file never becomes a hidden second source of truth that you cannot inspect. Terms already in the box are not added twice.

3.4 Between which ontologies

Two optional selectors, **From ontologies** and **To ontologies**, restrict the source and target of the mappings. Each option is annotated with how many mappings it takes part in. The  button between them swaps the two sides.

They compose with the term box in three useful ways:

- **Terms only** — every mapping involving those terms.
- **Terms + a target** — a bounded batch lookup: “map these 400 terms into MONDO”.
- **A source, and an empty term box** — the whole ontology: “everything DOID maps to”.

Pick exactly one source and the target list narrows to the ontologies actually reachable from it, with their counts, so you cannot pick a target that will return nothing.

3.5 Where should mappings come from?

Where should mappings come from?

Ontologies

Curated sets

Both

Everything OxO knows about.

The corpus control. The hint underneath changes with your choice.

This is the **Ontologies** / **Curated sets** distinction from § 1. **Both** is the default. Choose deliberately: a cross-reference an ontology asserts about its own terms and a mapping a curator drew between two projects are different kinds of evidence, and which you want depends on what you are doing.

THIS CONTROL DOES NOT HIDE INFERRED MAPPINGS

An inferred mapping chains premises from several sets at once, so it belongs to no single corpus and always survives this filter. Hiding inferences is a separate control, on the results table (§ 5.3). If choosing *Ontologies* leaves you with nothing but **SSSOM inference** rows, see § 11.

3.6 Order results by

Order results by

Best match ▼

Best match ranks by who asserts the mapping, then how strong it is.

Choice	Order	Use it when
Best match (default)	OxO's ranking — see § 6.	Always, unless you have a reason not to.
Highest confidence	By the mapping's declared confidence, descending.	You care only about the number the publisher put on it.
Most recent	By mapping date, descending.	You are checking what changed lately.

SORTING TURNS RANKING OFF

Any explicit sort — from this control or from a column header — *replaces* the relevance ranking rather than breaking ties within it. Sorting by confidence puts a confidence-0.95 inferred mapping above a confidence-0.9 ontology cross-reference. That is what you asked for; just be aware that you asked for it.

3.7 More options

Two settings sit behind the ► **More options** disclosure, because most searches never need them.

Label matching controls how free-text terms match, and nothing else — CURIEs and IRIs are always exact lookups.

- **Exact match (ignore case)** — the default. `cataract` finds *Cataract*.
- **Partial match** — substring-ish. Broad; expect volume.

- **Exact match (case-sensitive)** — for when case is meaningful.

Restrict the search to one or more mapping sets shows every mapping set Oxo holds, with its title, identifier, type and description. Click a row to toggle it. With nothing selected, all sets are searched. The count of selected sets is shown on the *More options* summary line even when it is collapsed, so a forgotten restriction cannot silently narrow your results.

4 Advanced search

The **Advanced** tab is a form of every SSSOM field Oxo indexes, grouped by Subject, Object, Predicate, Mapping, Authors/Reviewers, Provenance, Asserted mappings, and Other. Fill any combination; **all filled fields are combined with AND**.

Matching semantics depend on the field, and the placeholder text in each box tells you which:

- “*Exact match, case-sensitive*” — identifier-like fields (`subject_id` , `license` , ...).
- “*Text search*” — prose-like fields (`subject_label` , `comment` , ...).
- “*...(matches any list item)*” — the field holds a list; matching any one entry is enough.

WHAT ADVANCED DOES NOT HAVE

No corpus control, no *Order results by*, no from/to ontology selectors: Advanced is a flat, deterministic field query, sorted by subject identifier. Numeric and date fields (`confidence` , `mapping_date` , `distance` , ...) are not offered, because they need range inputs rather than a text box. Use Advanced to answer a precise question, and the Search tab to explore.

5 Reading the results

5.1 The columns

Column	Shows
Subject, Object	Identifier (bold), then label, then IRI. Each identifier and IRI has a copy button; subjects and objects also carry an <u>OLS</u> link that opens the term in a new tab.
Predicate	Same three lines, no OLS link. If the mapping carries a predicate modifier, a red NOT badge appears in front of it — that mapping asserts the <i>negation</i> . Do not skim past it.
Mapping justification	Why the mapping is claimed to hold — e.g. <code>semapv:ManualMappingCuration</code> .
Type	Asserted or SSSOM inference .
Mapping provider	Who supplied it.
Mapping set	The set’s title, with its identifier and a copy button beneath.

The eye icon at the end of a row opens the mapping detail page (§ 7).

5.2 One row can be several mappings

Two sources may assert the very same subject–predicate–object triple. Rather than repeat the row, Oxo collapses them into one **expandable** row. Such a row shows a badge per distinct type and a “*n* mappings” count; the triangle on the left expands it to the underlying mappings, each with its own justification, provider, set, and eye icon.

Where the collapsed members disagree on a field, the cell reads *Multiple* (or *Multiple sets*) rather than misleadingly showing one of them. If a triple has more members than the row can carry, a “**+*n* more — view all in Advanced search**” link takes you to the flat Advanced view filtered to exactly that triple.

5.3 Narrowing what you see

The **funnel icon** in a column header opens a filter with one box per underlying field — Subject offers *Subject ID*, *Subject label* and *Subject IRI* separately. Filters are “contains”, are combined with AND across every column, and apply about 400 ms after you stop typing. An active filter puts a dot on the funnel.

The funnel on the **Type** column is a different beast: a three-way choice of *All* (the default), *Asserted* only, or *SSSOM inference* only. **This is the control that hides inferred mappings** — not the corpus selector.

5.4 Re-ordering

The **sort icon** in a header lets you sort that column by identifier, label, or IRI, ascending or descending. Sorting is single-key *within* a column — choosing IRI replaces label — but multi-column *across* them: sort by Subject, then Object, and each icon shows a small number giving its place in the sort order. **Clear** removes that column from the sort.

Remember § 3.6: the moment you sort by anything, relevance ranking stops applying.

5.5 Paging and export

Page size is 5, 10 (the default), 20, 30, or 50 rows. The **download icon** in the toolbar exports the *entire* result set — not just the page you are looking at — as SSSOM-compliant TSV, named `oxo2-mappings.tsv`. Every filter you have applied is honoured by the export.

5.6 The address bar is the state

Page, page size, sort, type filter and every column filter live in the URL. Three consequences worth knowing:

- A results URL is **shareable** and **refresh-stable**. Send it to a colleague and they see what you see.
- Opening a mapping's details and pressing **Back** returns you to the same page, sort and filters — not to page 1.
- Defaults are omitted, so an untouched search has a clean URL.

6 How “Best match” ranks

OxO’s ranking rests on one editorial judgement: **trust provenance over predicate**. Who asserts a mapping tells you more about whether to believe it than how strongly the mapping is worded. An ontology that cross-references its own terms is staking its reputation on that statement; a `skos:exactMatch` in an inferred chain is a computation.

So four factors are applied, and they **multiply**. The gaps between provenance levels are wide enough that no combination of the other three can bridge them — the lower tiers order mappings *within* a provenance level, never across.

TIER 1 • PROVENANCE

Ontology ×10 000 › Curated ×1 000 › Inferred ×100, ÷5 per extra hop

decisive

TIER 2 • PREDICATE STRENGTH

owl identity ×2.0 › skos:exactMatch ×1.7 › closeMatch ×1.4 › broad/narrow ×1.2 › other ×1.0

TIER 3 • CURATION

Curated by a human ×1.3 › everything else ×1.0

TIER 4 • CONFIDENCE

×(1 + 0.3 × confidence). No declared confidence contributes ×1.0

tie-break

All four multiply together, then scale how well the term itself matched.

The arithmetic that makes tier 1 decisive is worth seeing once:

Mapping	Tier 1	Tier 2	Tier 3	Tier 4	Product
Ontology xref, weakest possible on every other tier	10 000	1.0	1.0	1.0	10 000
Curated mapping, <i>maxing out</i> every other tier	1 000	2.0	1.3	1.3	3 380
One-hop inference, maxing out every other tier	100	2.0	1.3	1.3	338

The best a curated mapping can do (3 380) still loses to the worst an ontology xref can do (10 000). The tiers are a ranking, not a blend.

Two consequences of the design that are easy to miss:

- **Logical identity outranks `skos:exactMatch`**. `owl:equivalentClass` is a formal claim with entailments; `skos:exactMatch` means “interchangeable in practice”. The former is the stronger statement, so it ranks above.
- **Recency is not a ranking factor**. Mapping dates are sparsely populated, and letting a date into the boost would silently reorder results on every query. It is offered as an explicit sort instead (§ 3.6), where you

can see that you chose it.

Ranking scales the score of how well your *term* matched, so a highly relevant inferred mapping can still appear above a barely-relevant ontology one. That is intentional: relevance to your query comes first, and provenance orders what is relevant.

7 The mapping detail page

Reached from the eye icon on any result row. It lays out everything SSSOM records about the mapping, in sections that hide themselves when empty — so what you see is what the source actually published:

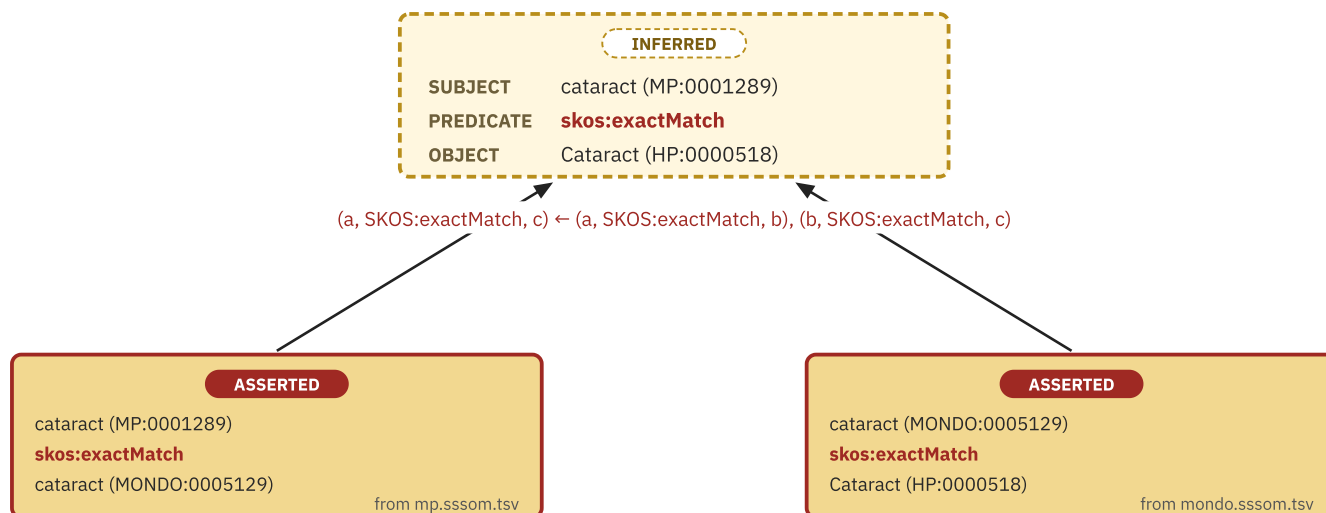
- **Mapping** — the triple, rendered large.
- **Inference Details** — for inferred mappings: the asserted mappings it rests on, and the explanation graph (§ 8).
- **Justification** — mapping justification and confidence.
- **Provenance** — author, creator, reviewer, mapping date, publication date, provider.
- **Mapping Set** — the set’s identifier, title, description, source, version.
- **Additional Information** — licence, comment, mapping tool and version.
- **Subject Details / Object Details** — category, type, source, source version.

A MAPPING DETAIL URL IS NOT A PERMALINK

This page renders the mapping handed to it by the results table. Open a `/mapping/<id>` URL directly, or press `F5` while on it, and you will get “*No mapping details available.*” To share a specific mapping, share the *results* URL that finds it — those are stable (§ 5.6).

8 Explaining an inferred mapping

When Oxo derived a mapping rather than reading it somewhere, it can show you exactly how. The detail page of an inferred mapping ends with a graph: **arrows point from premises to the conclusion**, so the mappings somebody actually asserted sit at the bottom and the derived one sits at the top.



Illustrative shape, not a real result. Real chains can be deeper and wider.

Reading the graph:

- **Solid box, dark red border, ASSERTED badge** — a mapping somebody published. Its bottom line names the mapping set it came from, which is how you see that a single inference crosses several sources.
- **Dashed box, amber border, INFERRED badge** — a mapping O_xO derived. The text below it names the SSSOM chaining rule that produced it, written as `conclusion ← premises`, with `a`, `b`, `c` standing for the terms being chained.
- Every arrow points **premise → conclusion**. Follow them upwards to see how the claim was built; follow them down to reach evidence a human wrote.

Drag to pan, and use the controls in the corner to zoom or fit. The nodes are diagrams, not links.

9 Inferred mapping sets

O_xO's inferences are themselves published as a mapping set with a resolvable identifier, so `/inferences` is a real page rather than a dead identifier. It shows the set's title, description, identifier, and the source sets it was derived from, followed by its mappings in the usual results table.

Unlike a normal search this page shows every inference type it holds, regardless of the default type filter — you navigated to a set of inferences, so hiding inferences would be perverse.

10 Glossary

Asserted mapping

A mapping somebody published, in an ontology or in a curated SSSOM file. Badged **Asserted**.

Inferred mapping (SSSOM inference)

A mapping O_xO derived by chaining asserted ones. Badged **SSSOM inference**. Always explainable (§ 8).

Mapping set

A published collection of mappings with its own identifier, title, licence and version.

Mapping set category

Whether a set is an *ontology's* own cross-references or a *curated* SSSOM file. Drives the corpus control (§ 3.5) and the top ranking tier (§ 6). It is an OxO judgement about the source, not something SSSOM records.

CURIE

A compact identifier: a prefix, a colon, a local name — `MP:0001289` . Expands to an IRI.

Distance

How many asserted mappings an inference had to chain. One hop is the strongest; each further hop divides the ranking by five.

Mapping justification

The SSSOM field saying *why* a mapping holds — hand-curated, lexically matched, logically reasoned, and so on.

Predicate modifier

A negation. A mapping carrying one asserts the triple does *not* hold; the table shows a red **NOT** badge.

SSSOM

The Simple Standard for Sharing Ontological Mappings — the vocabulary and TSV format OxO reads, stores and exports.

11 Things that will surprise you

1. **“Ontologies” or “Curated sets” returns only inferred mappings.** The corpus of a mapping set is recorded on each mapping when the data is loaded. If your OxO instance was indexed before that field existed, no asserted mapping carries a corpus, so filtering on one excludes them all — and only inferences, which are exempt from the filter, come back. **Both** (the default) is unaffected. A full data load fixes it; ask whoever runs the instance.
2. **Sorting a column silently disables relevance ranking.** There is no “sort within relevance”. See § 3.6.
3. **A `/mapping/<id>` URL does not survive a refresh.** Share the results URL instead. See § 7.
4. **The *Documentation* tab in the site header is a placeholder.** The document you are reading is the documentation.
5. **A red **NOT** badge inverts the row's meaning.** It is small and it is easy to skim past, and the mapping says the opposite of what the row appears to say.
6. **Advanced search ignores the corpus and sort controls.** They live on the Search tab only. See § 4.

OxO2 is developed by the Samples, Phenotypes and Ontologies Team (SPOT) at EMBL-EBI. Report a problem, or ask for a feature, at github.com/EBISPOT/oxo2/issues.

This document is generated from `docs/user/ui/oxo2-user-interface.html` in the OxO2 repository. The behaviour it describes is defined by the architectural decision records in `docs/adr/` — chiefly ADR-0011 (inference type), ADR-0013 (same-triple grouping), ADR-0024 (cross-ontology mapping), ADR-0026 (label matching) and ADR-0027 (mapping set category and provenance-led ranking).